

National Curriculum of Pakistan 2022-23

Assessment Framework

MATHEMATICS Grade-IX (SSC-I)

Details of Content Areas/ SLOs

Content Domain / Area	Topics	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods required (1 period= 40 minutes)
Domain A: Numbers and Algebra	Real Numbers	[SLO: M-09-A-01]: Explain, with examples, that civilizations throughout history have systematically studied living things [e.g., the history of numbers from Sumerians and its development to the present Arabic system.]	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	12
		[SLO: M-09-A-02]: Describe the set of real numbers as a combination of rational and irrational numbers.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-09-A-03]: Demonstrate and verify the properties of equality and inequality of real numbers.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-09-A-04]: Apply laws of indices to simplify radical expressions.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-09-A-05]: Express a number in scientific notations and vice versa.	Summative	U	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-06]: Describe logarithm of a number.	Formative	K	Question(s) will not be asked in the annual examination;	

					however, it will be part of regular teaching practice.	
		[SLO: M-09-A-07]: Differentiate between common and natural logarithm.	Summative	K	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-08]: Apply laws of logarithm to real life situations such as growth and decay, loudness of sound.	Summative	A	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-09]: Apply concepts of rational numbers to real word problems (such as inventory (stock taking), temperature, banking, measures of gain and loss, sources of income and expenditure).	Summative	A	Question(s) will be asked in the annual examination	
	Sets and Functions	[SLO: M-09-A-10]: Describe mathematics as the study of pattern, structure, and relationships.	Summative	U	Question(s) will be asked in the annual examination	15
		[SLO: M-09-A-11]: Identify sets and apply operations on three sets (Subsets, overlapping sets and disjoint sets), using Venn diagrams	Summative	A	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-12]: Solve problems on classification and cataloguing by using Venn diagrams for Scenarios involving two sets and three sets. Further application of sets.	Formative	U	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-09-A-13]: Verify and apply properties/laws of union and intersection of three sets through analytical and Venn diagram method.	Summative	A	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-14]: Apply concepts from set theory to real world problems (such as in demographic classification, categorizing products in shopping malls and music playlist by genre) Relation.	Summative	A	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-15]: Explain product, Binary Relations and its domain and range.	Summative	K	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-16]: Recognize that a relation can be represented by table, order pair and graphs.	Summative	A	Question(s) will be asked in the annual examination	

	Factorization	[SLO: M-09-A-17]: Identify common factors, trinomial factoring, concretely, pictorially and symbolically.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	15
		[SLO: M-09-A-18]: Factorize quadratic and cubic algebraic expressions. $a^4 + a^2b^2 + b^4$ or $a^4 + b^4$ $x^2 + px + q$ $ax^2 + bx + c$ $(ax^2 + bx + c)(ax^2 + bx + d) + k$ $(x + a)(x + b)(x + c) + k$ $(x + a)(x + b)(x + c)(x + d) + kx^2$ $a^3 + 3a^2b + 3ab^2 + b^3$ $a^3 - 3a^2b + 3ab^2 - b^3$ $a^3 \pm b^3$	Summative	U	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-19]: Find highest common factor and least common multiple of algebraic expressions and know relationship of LCM and HCF.	Summative	U/K	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-20]: Find square root of algebraic expression by factorization and division.	Summative	U/K	Question(s) will be asked in the annual examination	
		[SLO: M-09-A-21]: Apply the concepts of factorization of quadratic and cubic algebraic expressions to real world problems (such as engineering, physics, and finance.)	Summative	A	Question(s) will be asked in the annual examination	
	Linear Equations and Inequalities in one variable	[SLO: M-09-A-22]: Solve linear equations and inequalities with rational coefficients and represent the solution set on a real line.	Summative	U	Question(s) will be asked in the annual examination	10

		[SLO: M-10-A-23]: Solve two linear inequalities with two unknowns simultaneously	Formative	U	Question(s) will not be asked in the annual examination (This SLO will be taught in Class X)	
Domain B: Geometry	Coordinate Geometry	[SLO: M-09-B-01]: Derive distance formula by locating the position of two points in coordinate plane	Summative	U	Question(s) will be asked in the annual examination	15
		[SLO: M-09-B-02]: Calculate the midpoint of a line segment	Summative	K	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-03]: Find the gradient of a straight line when coordinates of two points are given.	Summative	K	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-04]: Find the equation of a straight line in the form $y = mx + c$	Summative	K	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-05]: Find the gradient of parallel and perpendicular lines.	Summative	U	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-06]: Apply distance and midpoint formulas to solve real life situations such as physical measurements or distances between locations.	Summative	A	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-07]: Apply concepts from coordinate Geometry to real world problems (such as, aviation and navigation, landscaping, map reading, longitude and latitude).	Summative	A	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-08]: Derive equation of a straight line in slope-intercept form, point-slope form, two-point form, intercepts form, symmetric form, normal form.	Summative	U	Question(s) will be asked in the annual examination	

		[SLO: M-09-B-09]: Show that a linear equation in two variables represents a straight line and reduce the general form of the equation of a straight line to the other standard forms.	Summative	U	Question(s) will be asked in the annual examination	
	Angle Between Lines	[SLO: M-09-B-10]: Find the angle between two coplanar intersecting straight lines.	Summative	K	Question(s) will be asked in the annual examination	07
		[SLO: M-09-B-11]: Find the equation of the family of lines passing through the point of intersection of two given lines.	Summative	U/K	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-12]: Calculate angles of the triangle when the slopes of the sides are given.	Summative	K	Question(s) will be asked in the annual examination	
	Logic	[SLO: M-09-B-13]: Differentiate between a mathematical statement and its proof.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	05
		[SLO: M-09-B-14]: Differentiate between an axiom, conjecture and theorem.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-09-B-15]: Formulate simple deductive proofs [algebraic proofs that require showing the LHS to be equal to the RHS. E.g., showing $(x - 3)^2 + 5 = x^2 - 6x$]	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
	Similar Figures	[SLO: M-09-B-16]: Identify similarity of polygons. Area and Volume of Similar Figures	Summative	K	Question(s) will be asked in the annual examination	05
		[SLO: M-09-B-17]: Solve problems using the relationship between areas of similar figures and volume of different solids	Summative	U	Question(s) will be asked in the annual examination	
	Geometrical Properties of regular	[SLO: M-09-B-18]: Solve real life problems that involve the properties of regular polygons, triangles	Summative	A	Question(s) will be asked in the annual examination	10

	Polygons, Triangles and Parallelograms	and parallelograms (such as building architectural structures, fencing, tiling, painting, carpeting a room).				
	Loci	[SLO: M-09-B-19]: Solve real life problems using the following loci and the method of intersecting loci for sets of points in two dimensions which are: at a given distance from a given point, at a given distance from a given straight line, equidistant from two given points equidistant from two given intersecting straight lines.	Summative	A	Question(s) will be asked in the annual examination	05
	Trigonometry	[SLO: M-09-B-20]: Identify angles in standard position, expressed in degrees and radians.	Summative	K	Question(s) will be asked in the annual examination	10
		[SLO: M-09-B-21]: Apply Pythagoras' theorem and the sine, cosine and tangent ratios for acute angles to find a side or of an angle of a right-angled triangle.	Summative	A	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-22]: Solve real life trigonometric problems in two dimensions involving angles of elevation and depression.	Summative	A	Question(s) will be asked in the annual examination	
	Trigonometric Identities	[SLO: M-09-B-23]: Prove the trigonometric identities and apply them to show different trigonometric relations.	Summative	K/A	Question(s) will be asked in the annual examination	05
		[SLO: M-09-B-24]: Solve real life problems involving trigonometric identities.	Summative	A	Question(s) will be asked in the annual examination	
	Bearing	[SLO: M-09-B-25]: Interpret and use three figure bearings.	Summative	K	Question(s) will be asked in the annual examination	07
		[SLO: M-09-B-26]: Solve problems involving bearing.	Summative	U	Question(s) will be asked in the annual examination	
		[SLO: M-09-B-27]: Apply the concepts of trigonometry	Summative	A	Question(s) will be asked in the annual examination	
	Construction of Triangle	[SLO: M-09-B-28]: Construct a triangle having given two sides and the included angle.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	10

		[SLO: M-09-B-29]: Construct a triangle having given one side and two of the angles.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-09-B-30]: Construct a triangle having given two of its sides and the angle opposite to one of them (with all the three possibilities).	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-09-B-31]: Draw angle bisectors, perpendicular bisectors, medians, altitudes of a given triangle and verify their concurrency.	Summative	U	Question(s) will be asked in the annual examination	
Domain C: Information Handling	Frequency Distribution	[SLO: M-09-C -01]: Construct a grouped frequency table, histogram (with unequal class intervals) and frequency polygon.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	05
	Measure of Central Tendency	[SLO: M-09-C -02]: Calculate the mean modal class and median of a grouped frequency distribution.	Summative	U	Question(s) will be asked in the annual examination	10
		[SLO: M-09-C -03]: Solve real life situations involving mean, weighted mean, median, and mode for given data (such as allocation of funds in different projects, forecasting future demographics, marketing, forecasting government budgets).	Summative	A	Question(s) will be asked in the annual examination	
	Probability	[SLO: M-09-C -04]: Calculate the probability of a single event and the probability of event not occurring	Summative	K	Question(s) will be asked in the annual examination	07
		[SLO: M-09-C -05]: Solve real life problems involving probability	Summative	A	Question(s) will be asked in the annual examination	
		[SLO: M-09-C -06]: Calculate relative frequency as an estimate of probability.	Summative	K	Question(s) will be asked in the annual examination	05

	Relative and expected frequencies	[SLO: M-09-C -07]: Calculate expected frequencies.	Summative	U/K	Question(s) will be asked in the annual examination	
		[SLO: M-09-C -08]: Solve real life problems involving relative and expected frequencies.	Summative	A	Question(s) will be asked in the annual examination	